Modeling and Analysis of High-Speed Through Silicon Via (TSV) Channel and Defects

Daniel H. Jung, Jonghoon J. Kim, Heegon Kim, Sumin Choi, Jaemin Lim, and Joungho Kim
Department of Electrical Engineering
Korea Advanced Institute of Science and Technology
Daejeon, South Korea
hyunsuk@kaist.ac.kr

Hyun-Cheol Bae, and Kwang-Seong Choi
IT Materials and Components Laboratory
Electronics and Telecommunications Research Institute
Daejeon, South Korea

***Abstract*—Through silicon via (TSV) based 3DIC has allowed vertical integration of multiple dies for wide I/O configuration. With thousands of TSVs, data transfer rate can be reduced, while maintaining the highest bandwidth compared to the systems in conventional integrated chips and packages. The challenges lie on high yield fabrication process. The trend in dimension of TSV is continuously decreasing, which also causes bumps and re-distribution layer (RDL) to be reduced for routing high number of I/Os. In this paper, we present the equivalent circuit models and analyze the TSV channels for investigation of the effect of possible defects. The verified models are used for characterizing the defects in TSV channel, and we validate the failure analysis method with electrical characteristic analysis in frequency-domain with S-parameter plots as well as time-domain waveforms with TDR.**

Keywords—Through silicon via (TSV), equivalent circuit model, failure analysis, defects

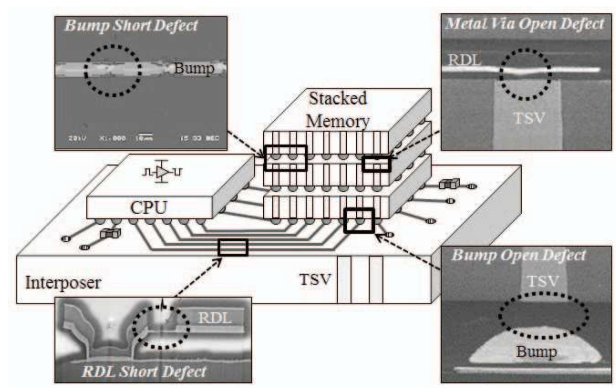


Fig. 1. Possible types and locations of defects in TSV based 3DIC. Open defects occur along the channel, normally between layers or between stacked dies.

I. Introduction

Through silicon via (TSV) based 3DIC has presented a new path to solve the ever increasing demand for wide I/O and high-speed system. By vertically stacking multiple chips, the length of interconnection is remarkably shortened, while the number of I/Os is increased up to the order of thousands [1]. In order to minimize the form factor and increase the bandwidth, the number of TSVs has to be maximized in a limited space. This causes continuous need for reduction of diameter and pitch of TSVs, which also decreases the pitch and size of bumps and re-distribution layers (RDLs). The reduced dimensions of interconnects cause challenges in fabrication process. Some of the main factors to consider for high yield fabrication includes, uniform bump size, copper filling without void, avoiding shorts between high pitch channels and alignment of stacked dies.

Subtle mistake may cause a variety of failures as shown in Fig. 1 [2]. Especially, open and short defects severely decrease the data transmission through the channels. The SEM images in Fig. 1 are short defect between bumps or RDLs formed between high pitch channels. Open defects normally occur between layers within a die by lack of copper filling in metal via or TSVs, or between dies caused by non-uniform height of bumps. Present failure analysis methods require destroying the test sample by grinding and analyzing the microscopic or SEM images of the cross section. The grinded samples experience a strong physical stress, which may cause additional failures. It is also very challenging to acquire a test sample that is grinded exactly at the area of interest. In order to establish advanced and reliable fabrication process for high yield products, non-invasive failure analysis is certainly required.

Non-invasive failure analysis methods are proposed to investigate TSV based 3DIC without destroying the test sample. In [3], [4], failures in TSV channels are analyzed by lock-in thermography (LIT) method. The works performs failure analysis for TSV channels by detecting the heat source caused by short defect. With known material properties of each layer, the thermal image pattern provides the layer information of the heat source. The LIT method limited to defects with heat source, such as short defect. The equivalent circuit models are presented in [2], [5]; however, the model with defect is not analyzed with dominant factors in each frequency range.

In this paper, the electrical performance of high-speed TSV channels are characterized by 3D FEM simulation and equivalent circuit modeling. Open defects are intentionally inserted between TSV in upper die and RDL on top of the bottom die by removing a bump and under bump metal (UBM) layer. The test structure is designed to have open defects in different locations to analyze its effect on the return loss in

978-1-5090-1442-2/16/$31.00 ©2016 IEEE

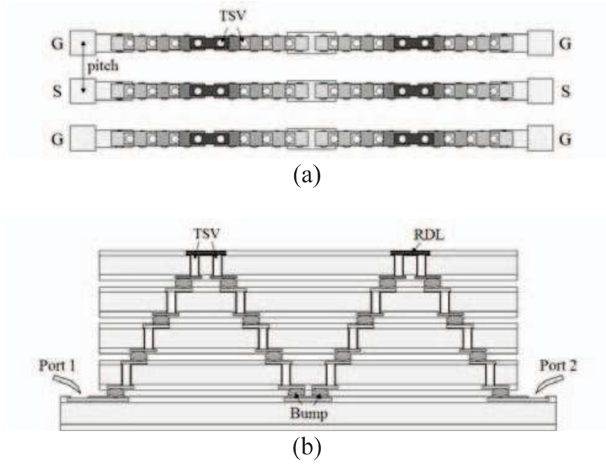


Fig. 2. GSG-type 5-layer daisy-chain structure for electrical characteristic analysis of TSV channel. 16 TSVs link the channel from port 1 to port 2, traveling four stacked dies four times upward and downward. (a) top view and (b) cross-sectional view.

frequency-domain and on TDR for time-domain analysis. The results evidently prove that open defects can be detected and localized by the amount of capacitance the signal experiences as it approaches and returns from the open defect.

II. Test Structure Design

The illustrations in Fig. 2 are the top view and cross-sectional view of the designed test structure. The structure is designed as GSG-type 5-layer daisy-chain structure with TSVs in top 4 dies and single layer RDL on top of the bottom die for probing. From the left side of the figure, the signal travels through 16 TSVs, experiencing four of the stacked silicon substrate two times upward and two times downward. The dimensions of each component is listed in Table I. The height, which is the thickness of silicon substrate, and diameter of TSV are 100 µm and 20 µm, respectively. The pitch between ground channels and signal channels is 250 µm. The dimensions, as well as material properties of each component and layer, are applied to the equations to obtain RLGC values for equivalent circuit modeling and circuit simulation.

The circuit diagram in Fig. 3 representation of the test structure designed for analysis. Each set of RLGC is represents vertical and horizontal interconnects: probe pads, front and back side RDLs, bumps, and TSVs. Along the path of the signal, the interconnects are characterized as series resistance and inductance. The insulation layer for electrical isolation of TSVs from semi-conductive silicon substrate and the oxide layer between RDLs and silicon substrate are represented as shunt capacitance. The shunt capacitance and conductance are the coupling and loss of signals between signal and ground channels.

After building the 3D and circuit models for daisy-chain structure, defects are inserted in different locations. Fig. 4 shows the cross-section of open defect. In the signal channel, one of the bumps is removed to analyze the effect of open defect by the varied location. This is represented as capacitance and conductance in parallel, which are linked in series with the

TABLE I. Parameters in TSV Daisy-chain Structure and Dimensions

Parameter Description	Dimensions (µm)
TSV height	100
TSV diameter	20
TSV Insulator thickness	0.5
Pitch	250
Bump height	10
Bump diameter	40
RDL thickness	2
Silicon Substrate Thickness	100

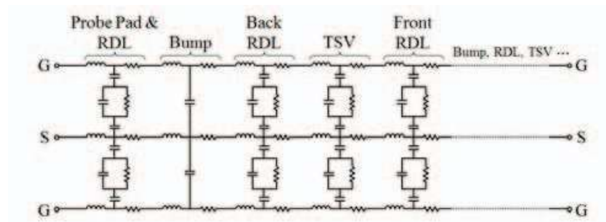


Fig. 3. Equivalent circuit model of GSG-type TSV channel. Each block represents front and back side RDLs, bumps, and TSVs. The sets of RLGC model are linked in series for circuit simulation.

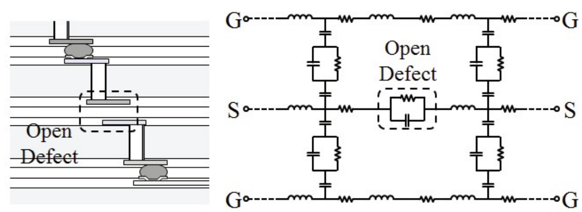
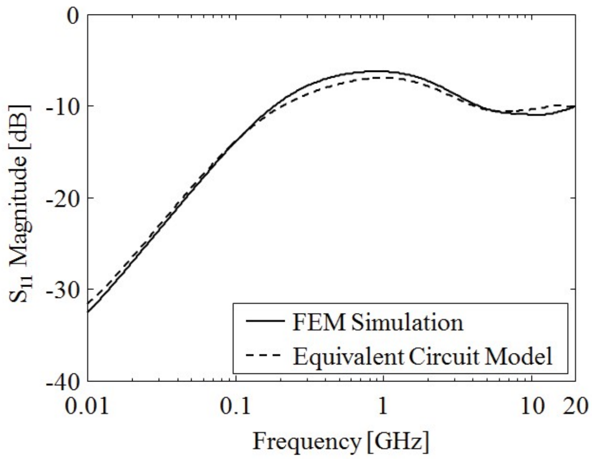


Fig. 4. The open defect structure and its equivalent circuit model. Open defect is designed by removing one of the bumps in signal channel. Since back side RDL of the upper die and front side RDL of the lower die are facing each other with dielectric passivation layer in between, the open defect can be represented as series capacitance. The signal loss also exists, therefore, conductance is connected in parallel with the capacitance

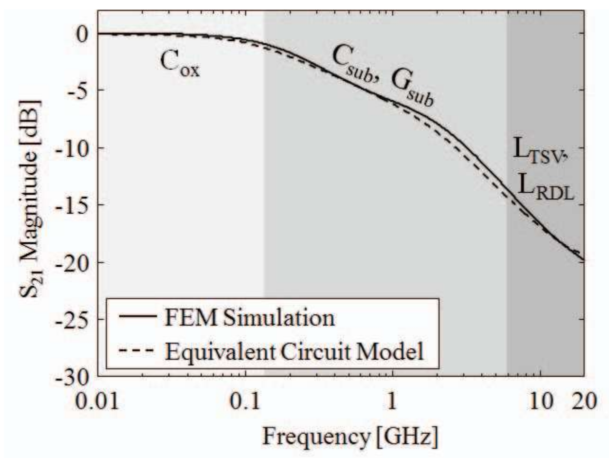
signal channel. As previously mentioned, the dimensions and material properties are applied to the equations from [6], [7] to calculate the values of each circuit component. The calculated values are applied in circuit simulation and S-parameter plots are extracted from HFSS, Ansys, for 3D FEM simulation and ADS, Keysight, for circuit model simulation.

III. Electrical Characteristic Analysis of TSV Channel

In order to analyze effect of defects on the electrical characteristic of TSV channel, S-parameters are extracted. Before inserting the defects in 3D and circuit models, the channels without defect is simulated for verification. Fig. 5 plots and compares S_{11} magnitude and S_{21} magnitude extracted from 3D FEM solver and circuit simulation of the model. The results prove that models are appropriately built to accurately predict the characteristics in all frequency ranges up to 20 GHz.



(a)



(b)

Fig. 5. S-parameter results obtained from FEM simulation and equivalent circuit model simulation. The results are compared in (a) S_{11} magnitude and (b) S_{21} magnitude plots, which both show the correlation. The verified model is used to simulate the TSV channels with model for open defect included.

After verifying and analyzing the channels without defect, open defect discussed in Fig. 4 is included for failure analysis in terms of electrical characteristics.

In S_{11} magnitude, the level of the curve is low in low frequency range, which shows that signal is well transmitted instead of reflected back to the port. As the frequency increases, the return loss increases due to existence of skin depth in higher frequency range, increasing the resistance of the channel. For S_{21} magnitude curve, each frequency range can be characterized by dominant factors from the circuit model. The capacitance formed between TSVs and silicon substrate is symbolized by C_{ox}. This value is relatively a large capacitance, which determines the range of flat region in low frequency range. As the frequency increases, the signal experiences the capacitance and conductance in silicon substrate between signal and ground TSVs. In higher frequency range, the inductance starts to dominate the curvature, and forms resonance. The resonant frequency is expected to be in higher frequency than the plotted, however, the left side of the expected valley is shown.

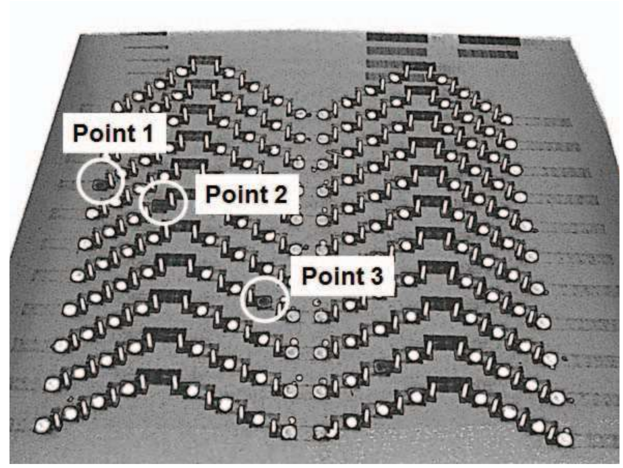
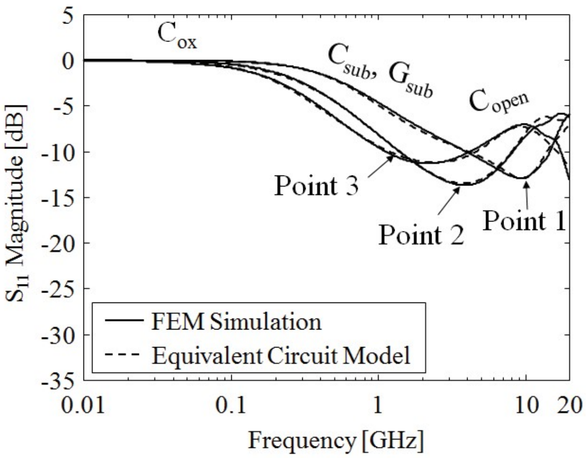


Fig. 6. 3D CT image of the test structure. Bumps in three labeled points are removed to analye the effect of open defect on the channel by its distance from the ports. The signal travels through 0, 3, and 7 TSVs as it approaches point 1, point 2, and point 3, respectively.

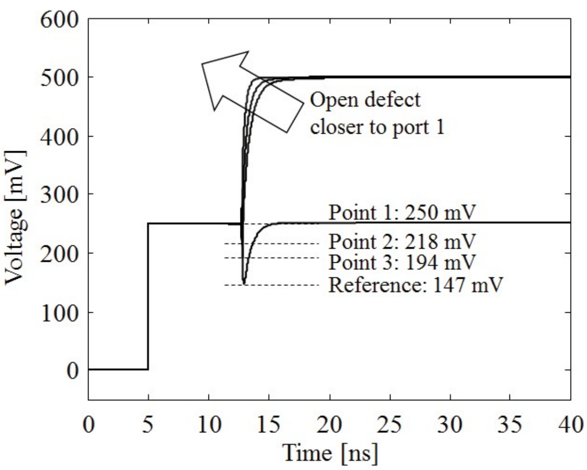
With the verified model, the defect model in Fig. 4 is inserted for simulation. As shown in the image in Fig. 6, the daisy-chain structures are designed to be in parallel with 250 μm apart from the adjacent channels. At point 1, point 2, and point 3, the bumps are removed to have open defect in each channel. Port 1 is located at the left side of the image, where S_{11} is measured. For channels with defects, only return loss curves show the difference according to the location; S_{21} curve shows similar characteristics for same type of defects and does not depend on the location. The S_{11} magnitude plots obtained from FEM simulation and circuit model are plotted in Fig. 7(a). The presented model with defect is also verified by comparing S_{11} magnitudes with the results from FEM simulation.

The dominant factors are labeled for each frequency range. The S_{11} plot with open defect shows similar characteristics as the S_{21} plot without the defect. In low frequency range, the signal experiences capacitance of higher value and starts on a down slope when capacitance in silicon substrate takes the dominance. As the signal approaches the open, the number of TSVs traveled determines the total amount of C_{ox}, therefore, the flat region is wider for the channel with the defect closer to port 1. The curvature clearly shows that the flat region is wider for channel with defect at point 1. In a similar manner, the signal traveling through lower silicon capacitance and conductance results in higher level of S_{11} magnitude. In higher frequency range, the value of C_{open}, which is the series capacitance, determines the level. Depending on the location of open defect, the number of TSVs the signal travels determines the S_{11} curvature; by comparing the results with reference in Fig. 5(b), the defect can be localized without destroying the test sample.

The waveforms in Fig. 7(b) are the TDR results from FEM simulation. The reflected signal is analyzed investigate the effect of open defect on the channel in time-domain. Since the



(a)



(b)

Fig. 7. Simulation results of TSV channel with open defects at point 1, point 2, and point 3. (a) S_{11} magnitudes from 3D FEM simulation and the equivalent circuit model. (b) TDR waveforms from 3D FEM simulation. Both plots show the characteristic change according to the locatino of defect in TSV channel.

channel is too short to determine the location of open defect by time delay, the capacitance information is used for the failure analysis. As previously discussed, the amount of capacitance the signal experiences dominantly affects the electrical characteristics. Point 1 is the closest location of the defect, therefore, the shunt capacitance is the least; the reference TDR waveform is the result from the channel without defect, in which all TSVs are linked in series. The capacitive peak is the largest for the reference, and decreases as the open defect moves closer to port 1. In addition to the peak value, the rising time after the peak is also determined by the capacitance; with the smallest capacitance, open at point 1 shows the highest rising time. Since port 2 terminates the channel, the waveform from the reference returns to 250 mV after the capacitive peak. With analysis of both S-parameter and TDR results, the open defect in TSV channel can be detected and located

IV. Conclusion

In this paper, 5-layer GSG-type daisy-chain structure is analyzed to see the effect of open defect on the TSV channel. The equivalent circuit model is built and simulated. By comparing the S-parameter results from circuit simulation with that from FEM simulation, the accuracy of the model is verified for channels with and without open defect. The dominant factors in each frequency range are determined by analyzing the equivalent circuit model. With 3D model and circuit model of the test structure, S-parameter and TDR waveforms are extracted. Both results clearly show the change in electrical characteristics of the channel depending on the location of defects. With precise reference of the TSV channel, the open defect can be accurately detected and located without destroying the test sample.

Acknowledgment

We would like to acknowledge the technical support from ANSYS (ANSYS HFSS) and the financial support from the R&D Convergence Program of MSIP (Ministry of Science, ICT and Future Planning) and ISTK (Korea Research Council for Industrial Science and Technology) of Republic of Korea (Grant B551179-12-04-00) and International Collaborative R&D Program (funded by the Ministry of Trade, Industry and Energy (MKE, Korea) [N0000899, Glass interposer based RF FEM for Next Generation Mobile Smart Phone]

References

[1] A. Papanikolaou, D. Soudris and R. Radojcic, Three Dimensional System Integration, Springer: New York, 2011.

[2] Jung, D.H.; Heegon Kim; Kim, J.J.; Sukjin Kim; Joungho Kim; Hyun-Cheol Bae; Kwang-Seong Choi, "Modeling and analysis of defects in through silicon via channel for non-invasive fault isolation," in 3D Systems Integration Conference (3DIC), 2015 International , vol., no., pp.TS8.29.1-TS8.29.4, Aug. 31 2015-Sept. 2 2015

[3] Schmidt, C.; Wadhwa, K.; Reverdy, A.; Reinders, E., "Localization of electrical active defects caused by reliability-related failure mechanism by the application of Lock-in Thermography," Reliability Physics Symposium (IRPS), 2013 IEEE International , vol., no., pp.5B.4.1,5B.4.6, 14-18 April 2013

[4] Krause, M., et al., "Characterization and failure analysis of TSV interconnects: From non-destructive defect localization to material analysis with nanometer resolution," Electronic Components and Technology Conference (ECTC), 2011 IEEE 61st , vol., no., pp.1452,1458, May 31 2011-June 3 2011

[5] Jung, D.H.; Heegon Kim; Kim, J.J.; Sukjin Kim; Joungho Kim; Hyun-Cheol Bae; Kwang-Seong Choi, "Fault detection and isolation of multiple defects in through silicon via (TSV) channel," in 3D Systems Integration Conference (3DIC), 2014 International , vol., no., pp.1-5, 1-3 Dec. 2014

[6] Joohee Kim, "High-Frequency Scalable Electrical Model and Analysis of a Through Silicon Via (TSV)," Components, Packaging and Manufacturing Technology, IEEE Transactions on , vol.1, no.2, pp.181,195, Feb. 2011

[7] S. H. Hall, *High Speed digital* System Design. New York: Wiley.